American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Junior Division

1. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$A\,\overline{B} + \overline{A\,B} + A$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A\,B} \; + \; \overline{A(B + \overline{A})}$$

A. $\overline{A} + \overline{B}$
B. $A + \overline{A} * \overline{B}$
C. $\overline{A} + B$
D. $A + B$
E. $\overline{A} + A\overline{B}$

3. Data Structures

How many nodes have only 1 child in the binary search tree for:

EQUESTRIAN

A. 1
B. 2
C. 3
D. 4
E. 5

4. Data Structures

Given an initially empty queue and the following commands on the queue, what items are left from the rear to the front of the queue?

```
PUSH("F")
PUSH("E")
PUSH("N")
x = POP()
PUSH("C")
x = POP()
PUSH("I")
PUSH("N")
x = POP()
PUSH("G")
x = POP()
x = POP()
```

A. E F
B. N G
C. G F
D. G N
E. I N

5. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array *arr*? Note that arr(0) has a value of 2.

2	3	4	7	6	5	1

```
a = arr(3) - arr(5)
b = arr(6) + arr(1)
output arr(a) * arr(b)
```

A. 18
B. 24
C. 20
D. 12
E. 14

6. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array *arr?* Note that arr(2,3) has a value of 0.

3	6	7	1	4
1	5	2	3	3
7	4	6	0	3

```
s = 0
for x = 1 to 2
     for y = 0 to 4
          if arr(x,y) < arr(x-1,y) then
               s = s + 1
          end if
     next y
next x
output s
```

A. 6
B. 4
C. 5
D. 3
E. 7

American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Junior Division

1. 布尔代数

请问有多少个有序对可以使得下列布尔表达式的结果为真？

$$A\,\overline{B} + \overline{A\,B} + A$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. 布尔代数

请简化下列布尔表达式：

$$\overline{A\,B} \;+\; \overline{A(B + \overline{A})}$$

A. $\overline{A} + \overline{B}$
B. $A + \overline{A} * \overline{B}$
C. $\overline{A} + B$
D. $A + B$
E. $\overline{A} + A\overline{B}$

3. 数据结构

请问在以下字符串构成的二叉搜索树中，有多少个节点只有一个子节点？

EQUESTRIAN

A. 1
B. 2
C. 3
D. 4
E. 5

4. 数据结构

给定一个初始为空的队列，以及**下列**对**该**队列的操作**指令，请问从队尾到队首依次剩下哪些元素？**

```
PUSH("F")
PUSH("E")
PUSH("N")
x = POP()
PUSH("C")
x = POP()
PUSH("I")
PUSH("N")
x = POP()
PUSH("G")
x = POP()
x = POP()
```

A. E F
B. N G
C. G F
D. G N
E. I N

5. 这个程序是做什么的？(数组)

已知 arr(0) 的值为 2， 执行下列数组 *arr*，请问此程序的输出值是多少？

2	3	4	7	6	5	1

```
a = arr(3) - arr(5)
b = arr(6) + arr(1)
output arr(a) * arr(b)
```

A. 18
B. 24
C. 20
D. 12
E. 14

6. 这个程序是做什么的？(数组)

已知 arr(2,3) 的值为 0， 执行下列数组 *arr*，请问此程序的输出值是多少？

3	6	7	1	4
1	5	2	3	3
7	4	6	0	3

```
s = 0
for x = 1 to 2
     for y = 0 to 4
          if arr(x,y) < arr(x-1,y) then
               s = s + 1
          end if
     next y
next x
output s
```

A. 6
B. 4
C. 5
D. 3
E. 7